



[AGU: Earth Surface]

Supporting Information for

[A Global Empirical Model for Near Real-time Assessment of Seismically-induced Landslides]

[M.A. (Nowicki) Jessee, M.W. Hamburger, K. Allstadt, D.J. Wald, S. Robeson, H. Tanyas, M. Hearne, E.M. Thompson]

[Indiana University, Indiana University, U.S.G.S., U.S.G.S., Indiana University, University of Twente, U.S.G.S., U.S.G.S.]

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Additional Supporting Information (Files uploaded separately)

Caption for Table S3 (table uploaded as separate file)

Introduction

This document of supporting information includes full figures, whereas a few key examples were chosen and highlighted in the main text. For example, in the manuscript we show a few examples of landslide probability maps for individual earthquakes in comparison to their mapped landslide inventories. In this supplement, we provide the landslide probability maps for each of the training earthquakes as well as the blind validation earthquakes. This document includes additional figures and an additional table to aid in understanding the analysis described in the main text. This document also includes tables of model comparisons.

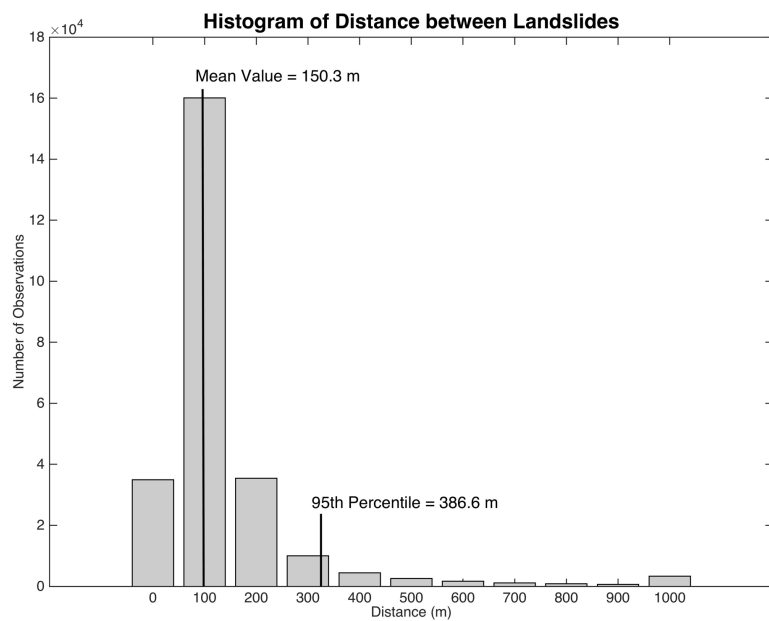
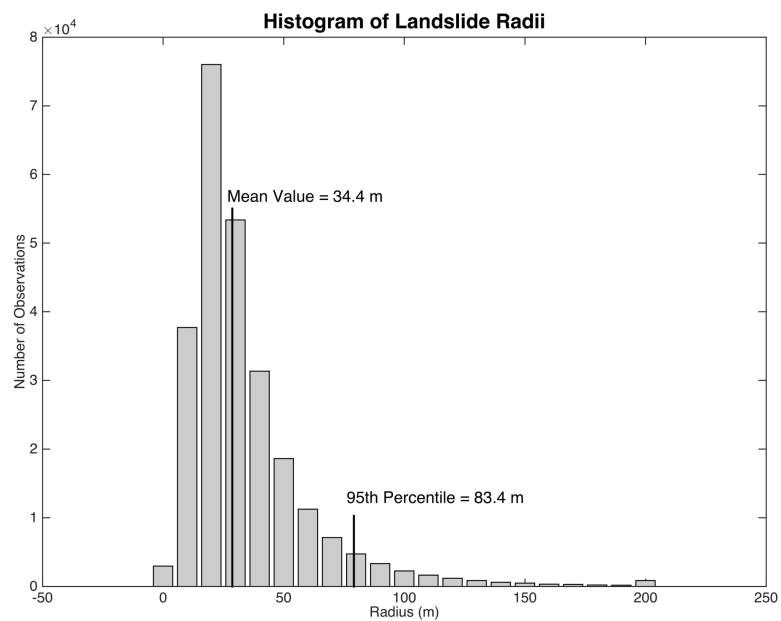
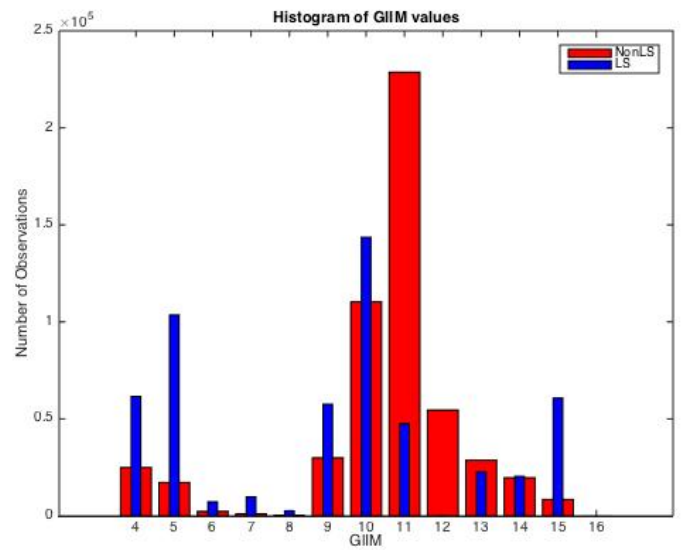
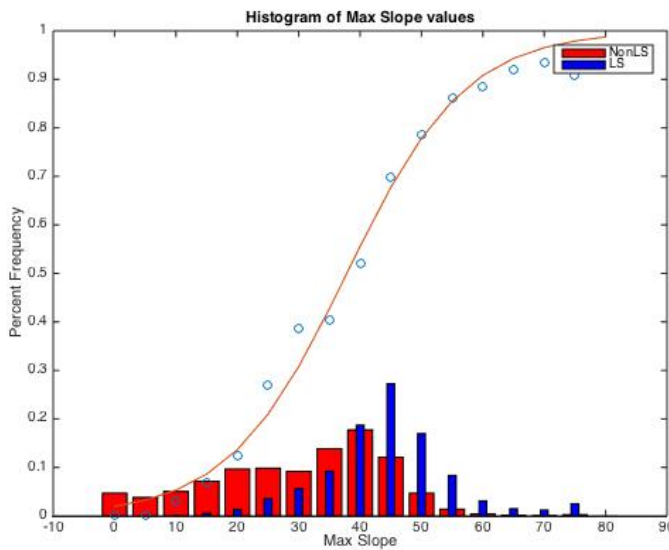
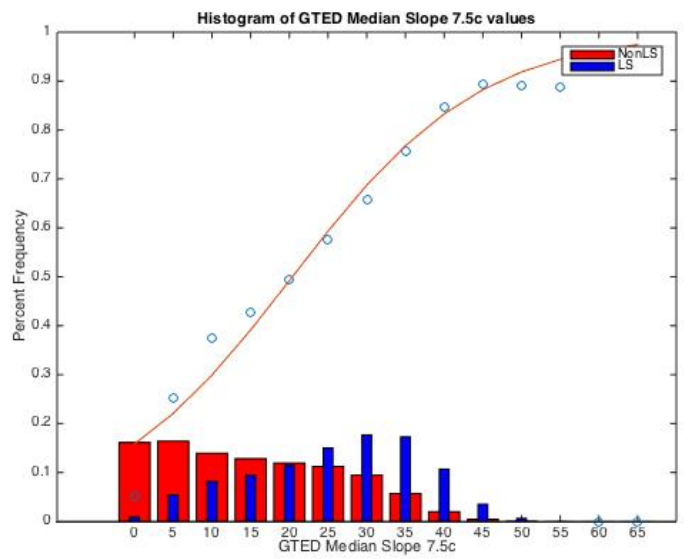
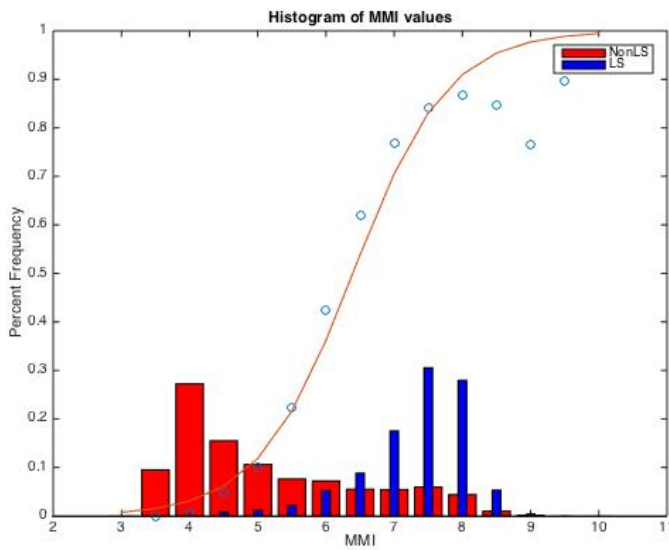
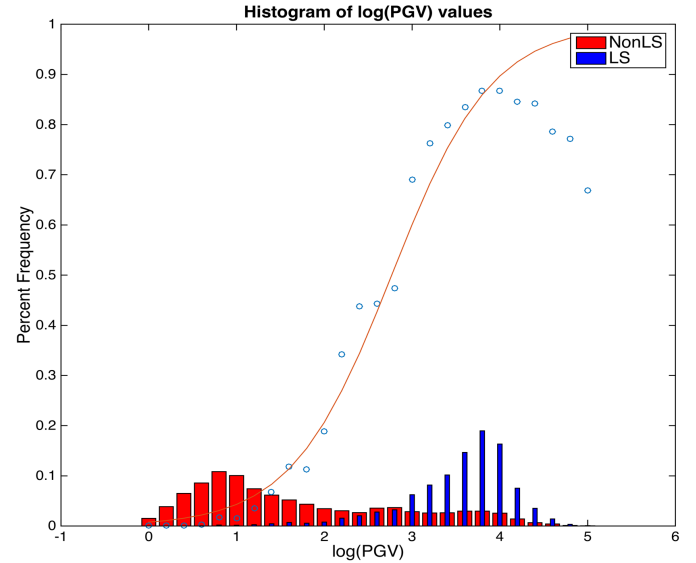
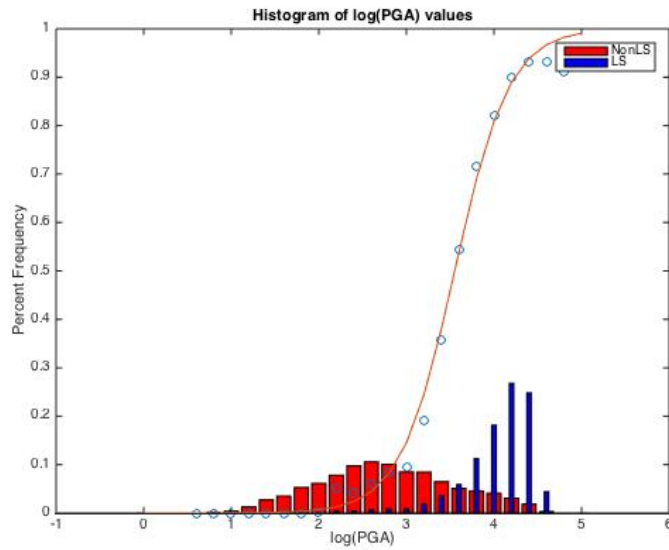
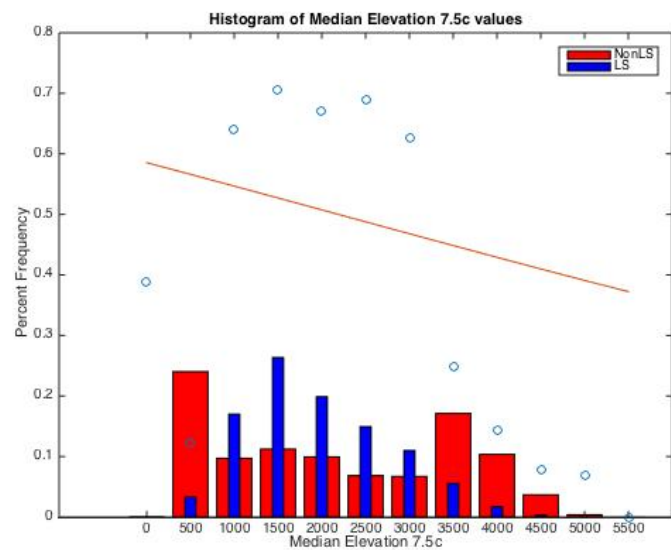
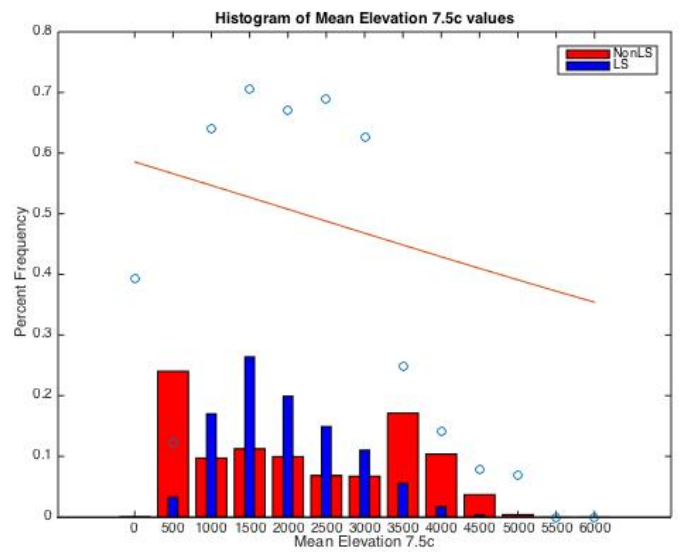
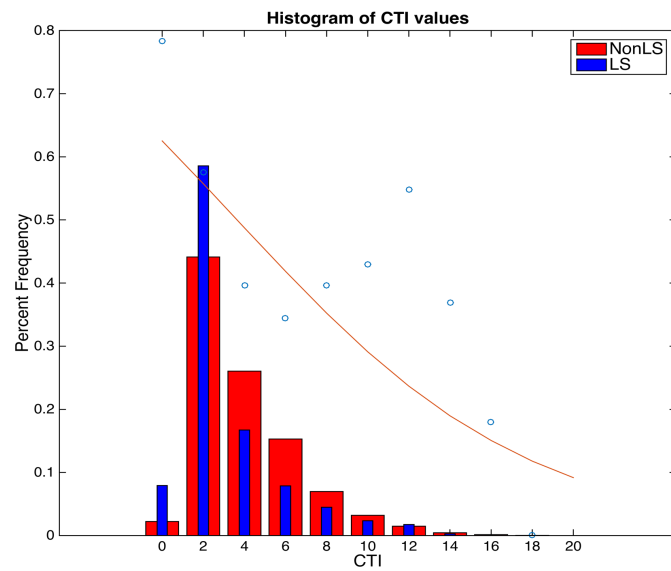
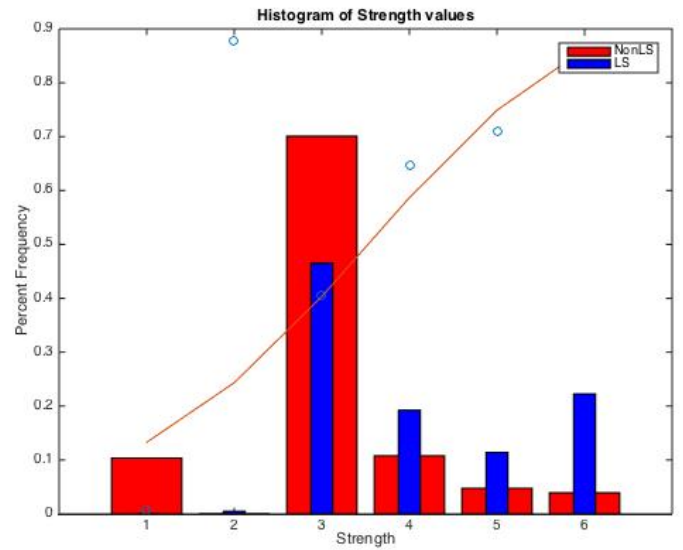
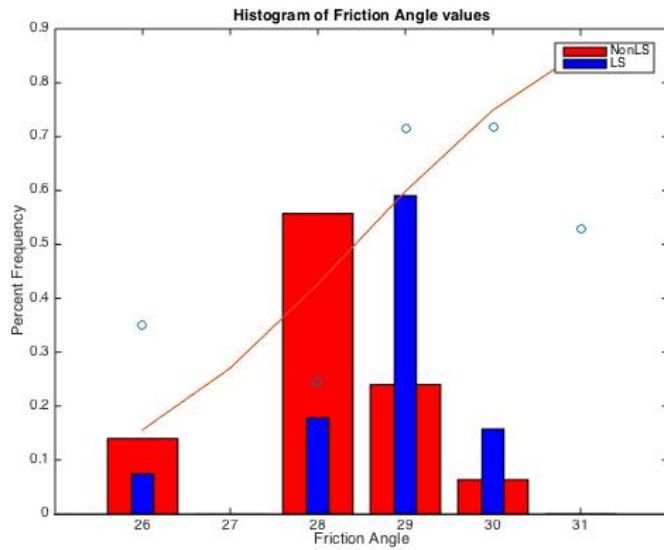
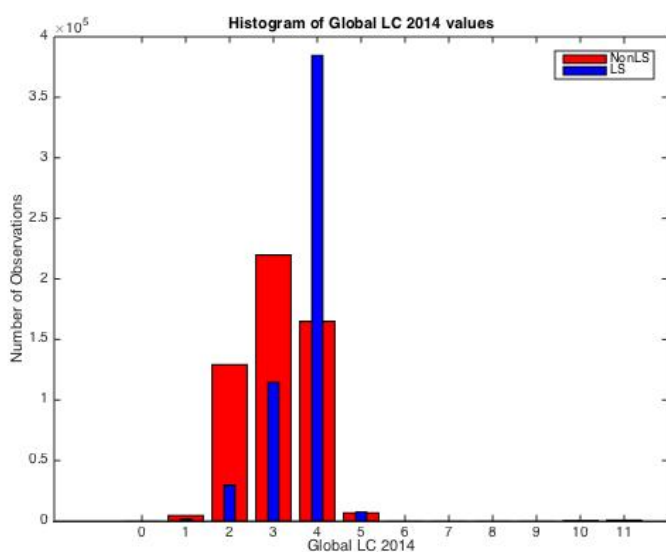
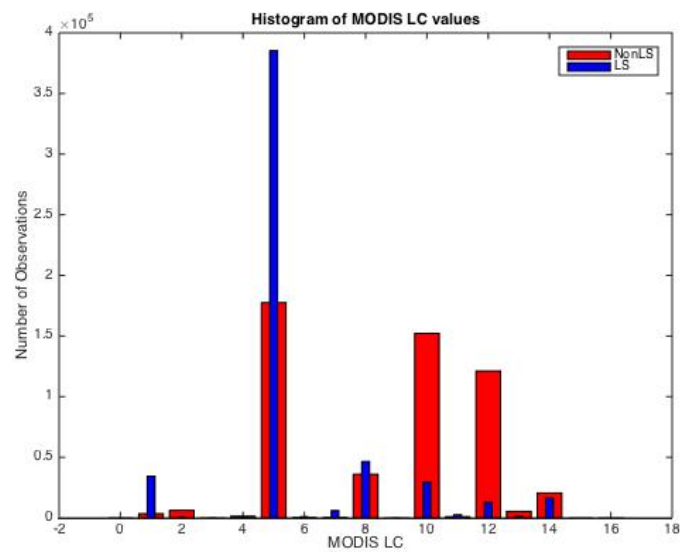
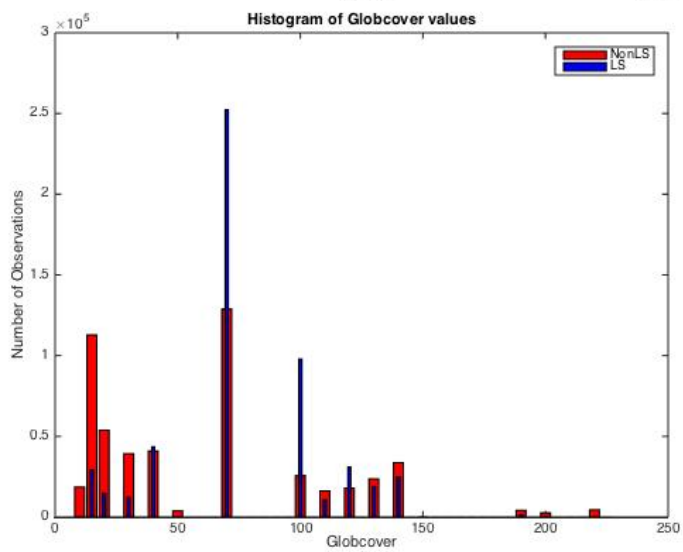
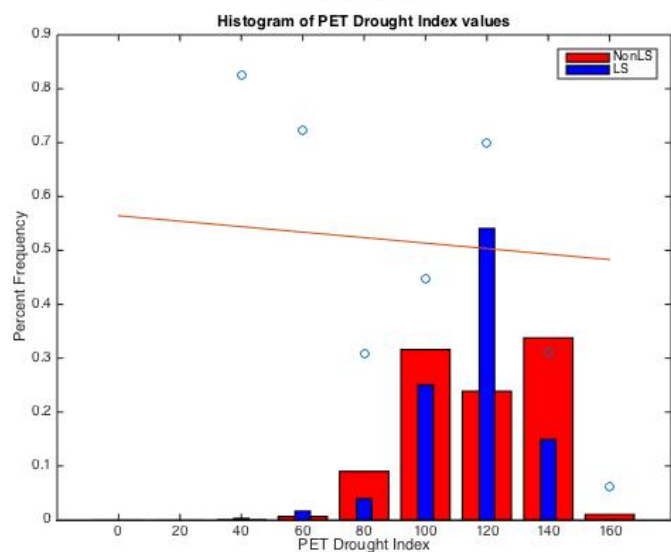
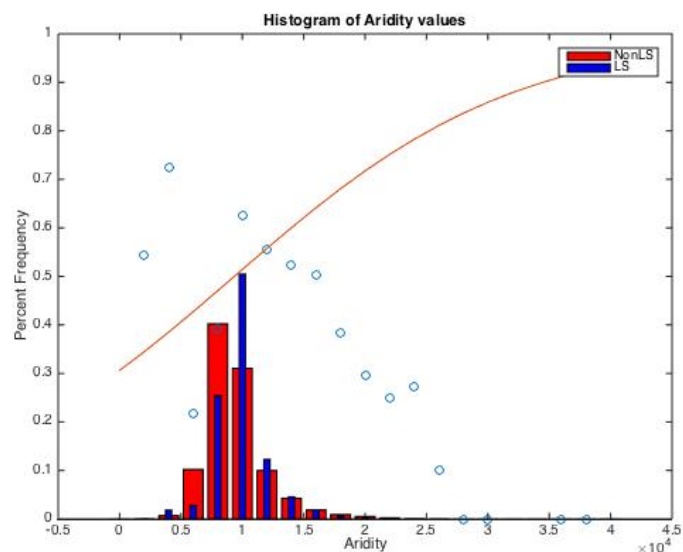
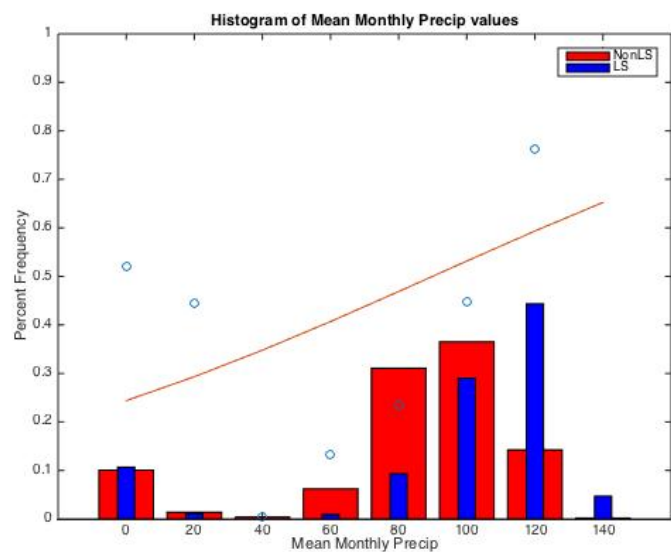


Figure S1. Histograms of data used to choose buffer sizes.







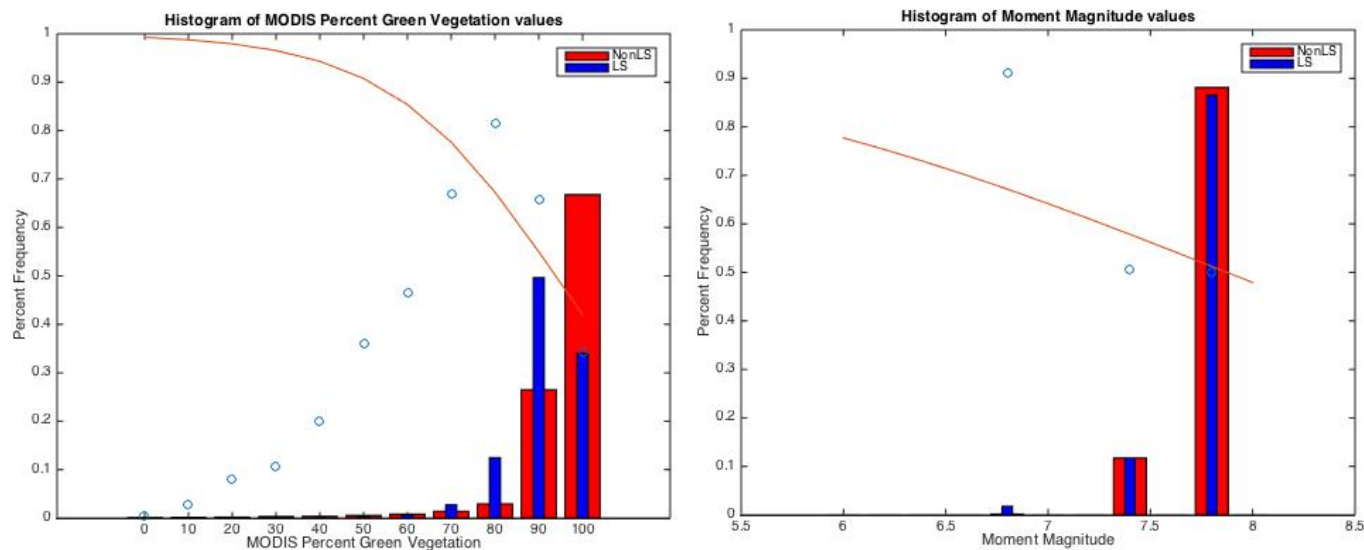


Figure S2. Example histograms of predictor variables tested within the global landslide model. Note the difference in observed values for landslide versus non-landslide observations (blue and red bars, respectively). Red lines show a logistic regression model between the individual variable and landslide/non-landslide observations. Blue circles show the fraction of observations that had a landslide in each bar.

Figure S3A. Guatemala

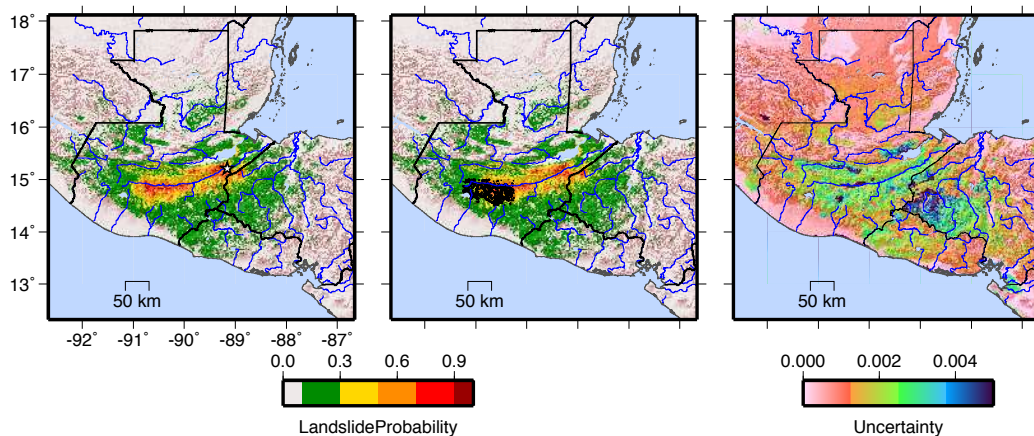


Figure S3B. Friuli, Italy

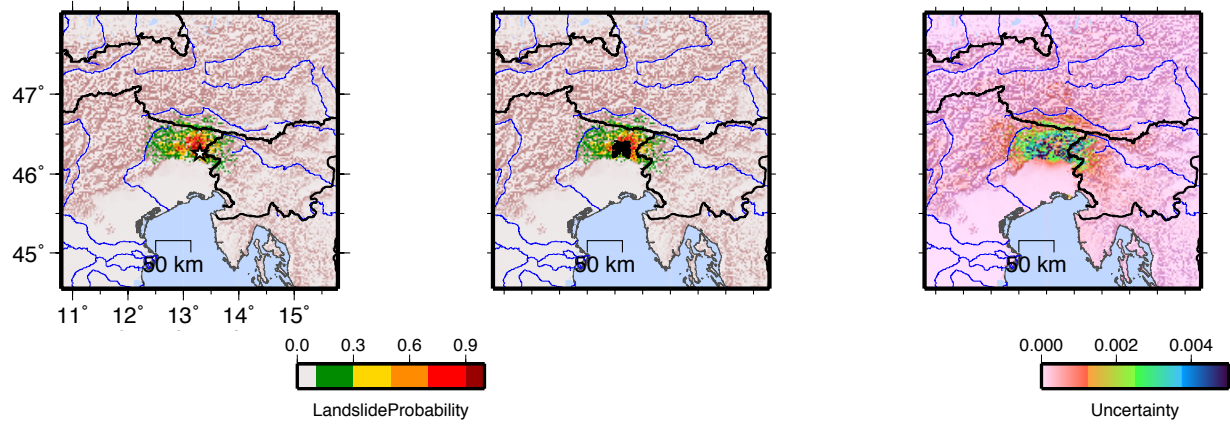


Figure S3C. Coalinga, California

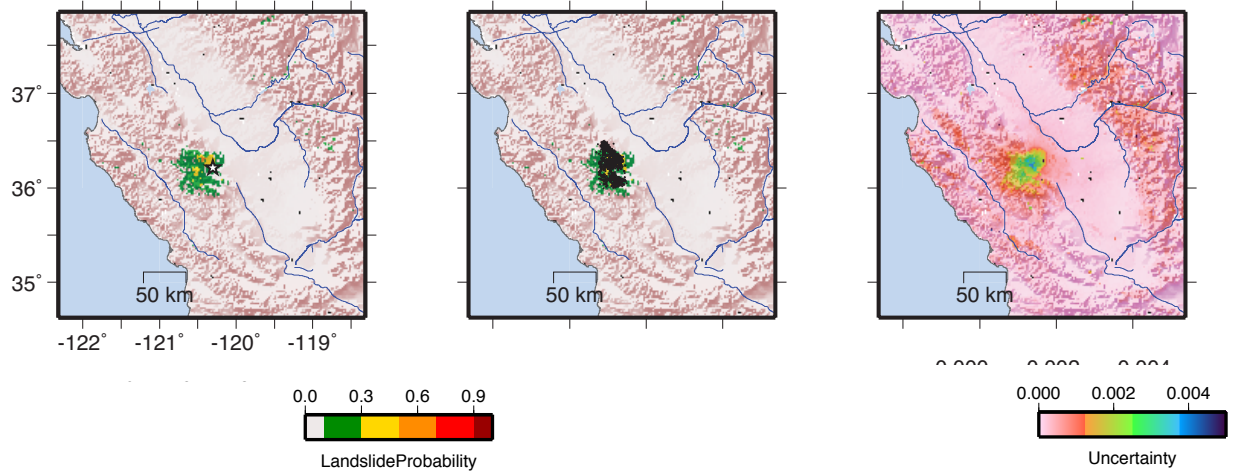


Figure S3D. Northridge, California

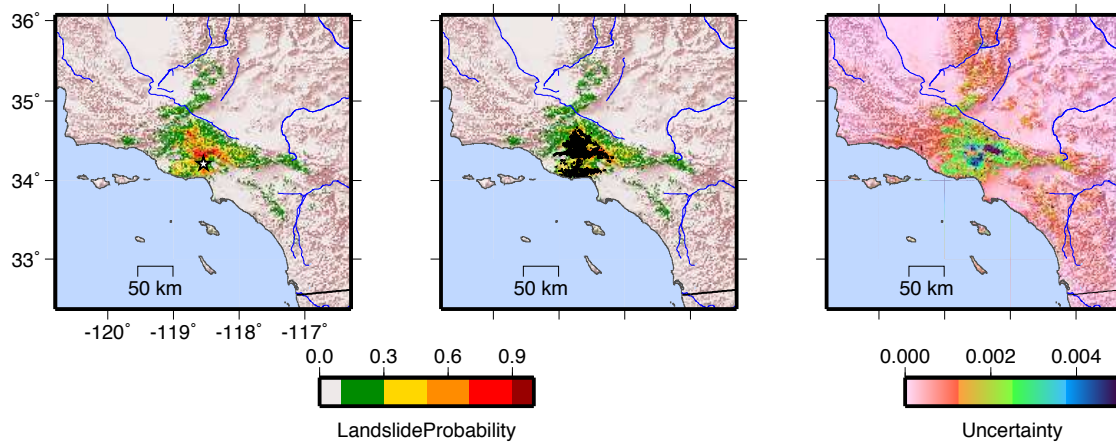


Figure S3E. Denali, Alaska

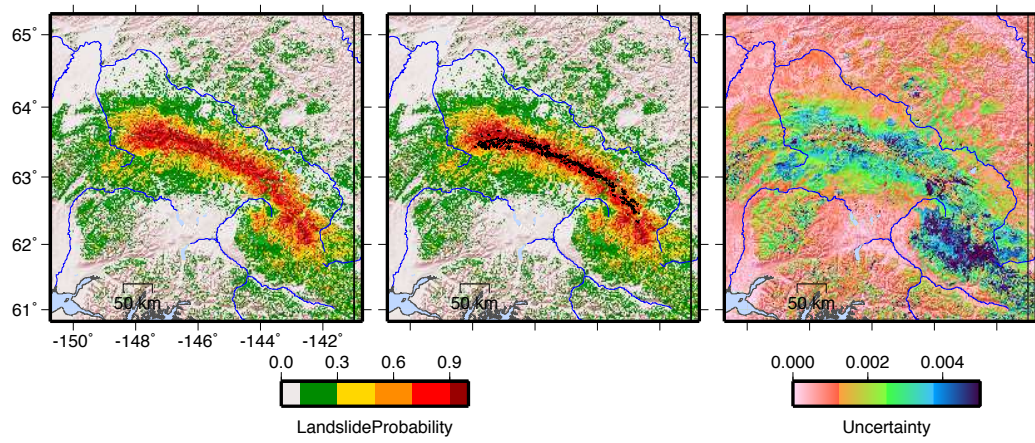


Figure S3F. Lefkada, Greece

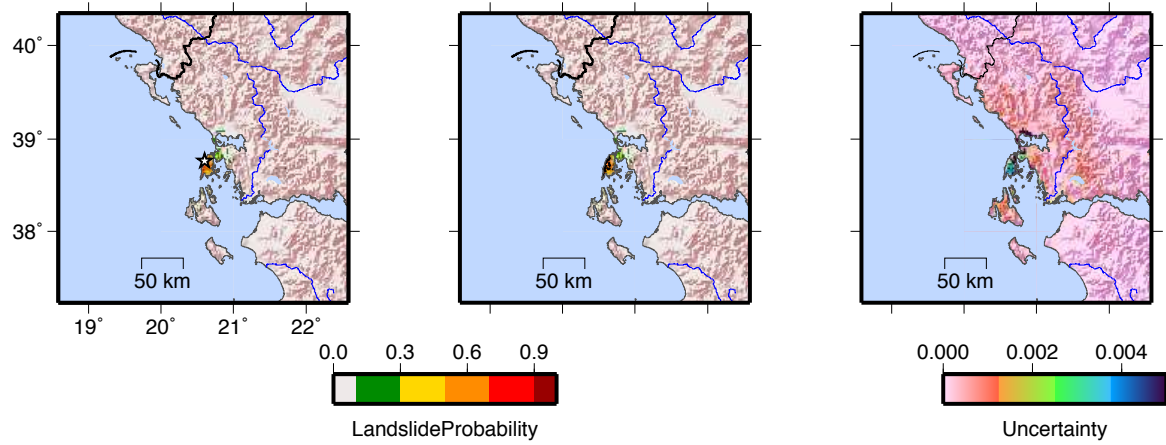


Figure S3G. Niigata-Chuetsu, Japan

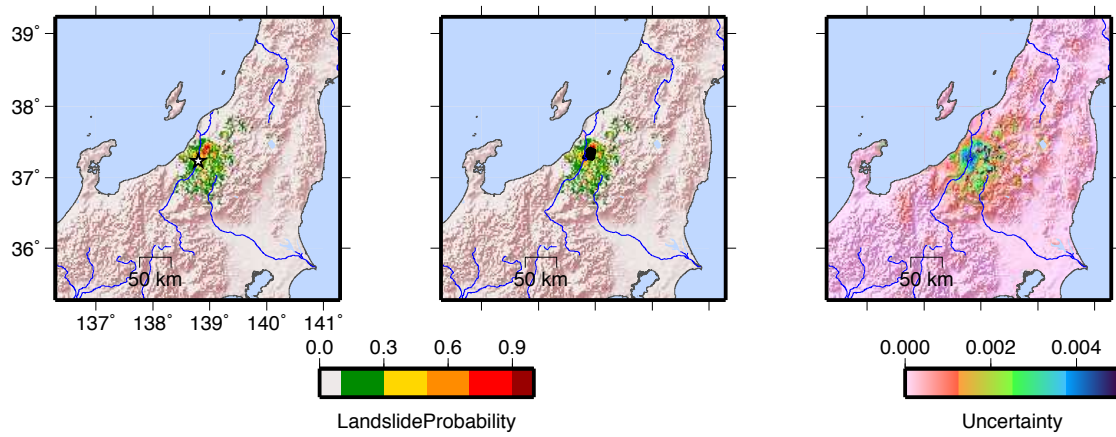


Figure S3H. Kashmir, Pakistan

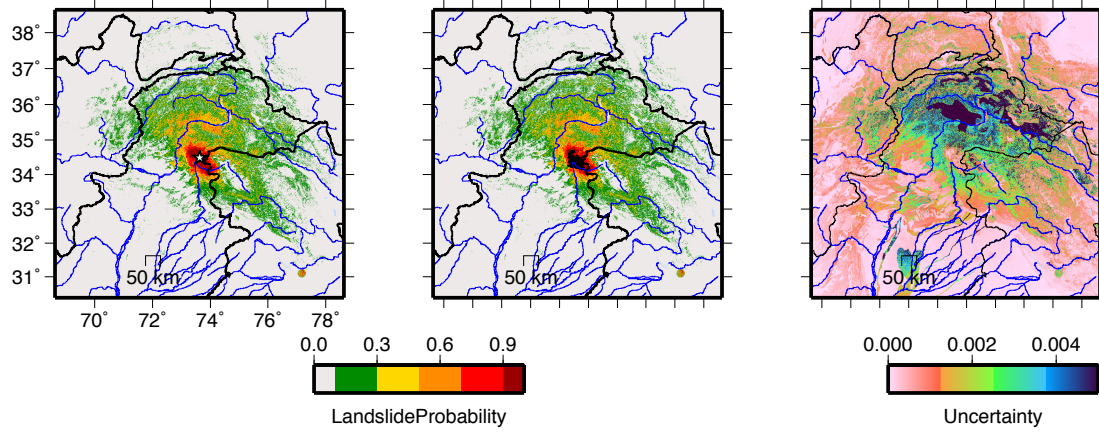


Figure S3I. Kiholo Bay, Hawaii

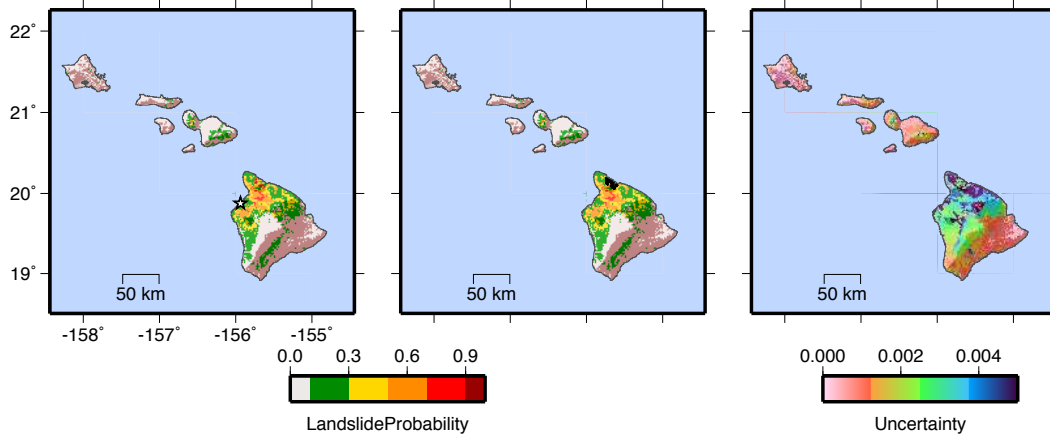


Figure S3J. Niigata-Chuetsu-oki, Japan

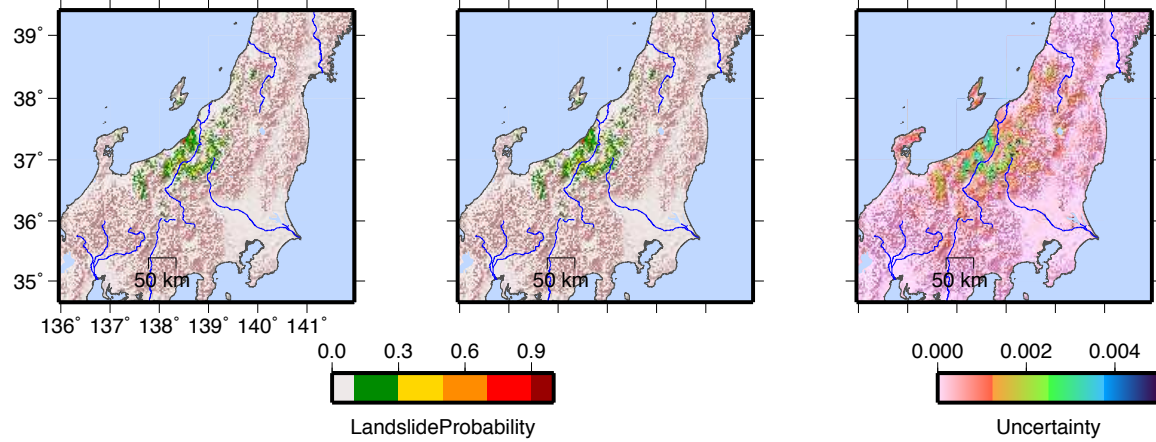


Figure S3K. Aysen Fjord, Chile

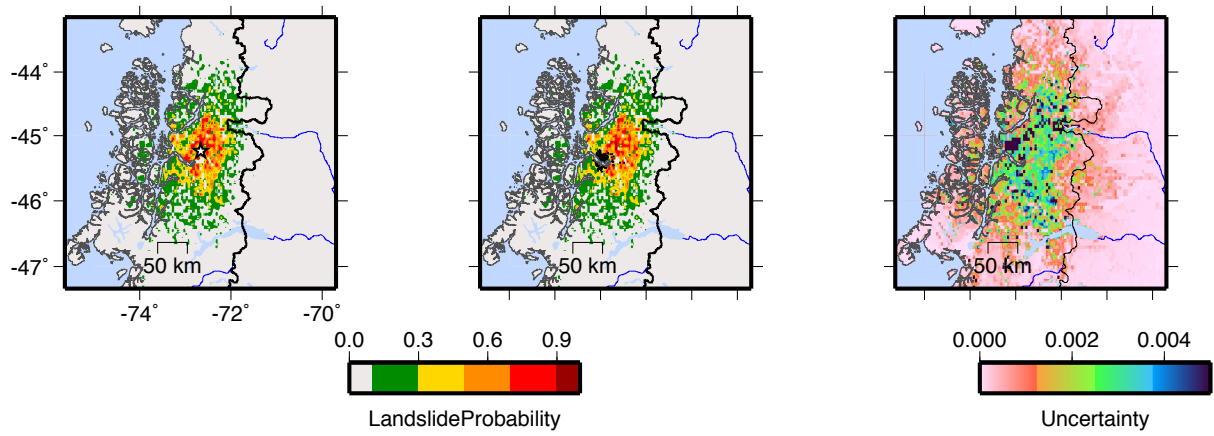


Figure S3L. Iwate-Miyagi-Nairuku, Japan

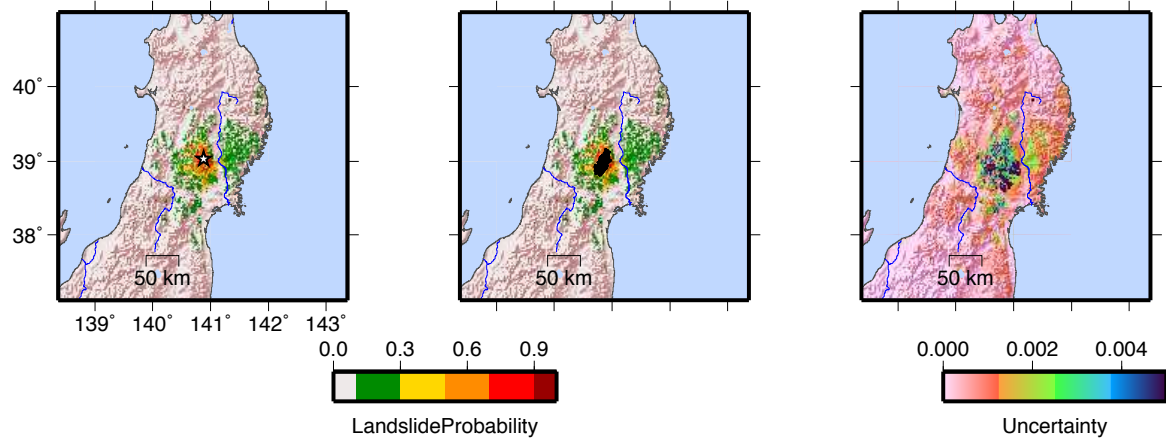


Figure S3M. Sierra Cucapah, Mexico

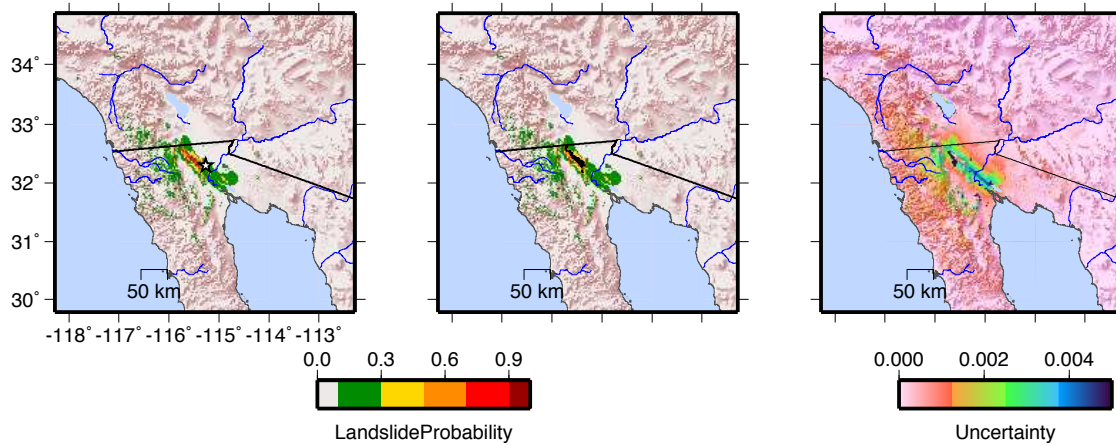


Figure S3N. Yushu, China

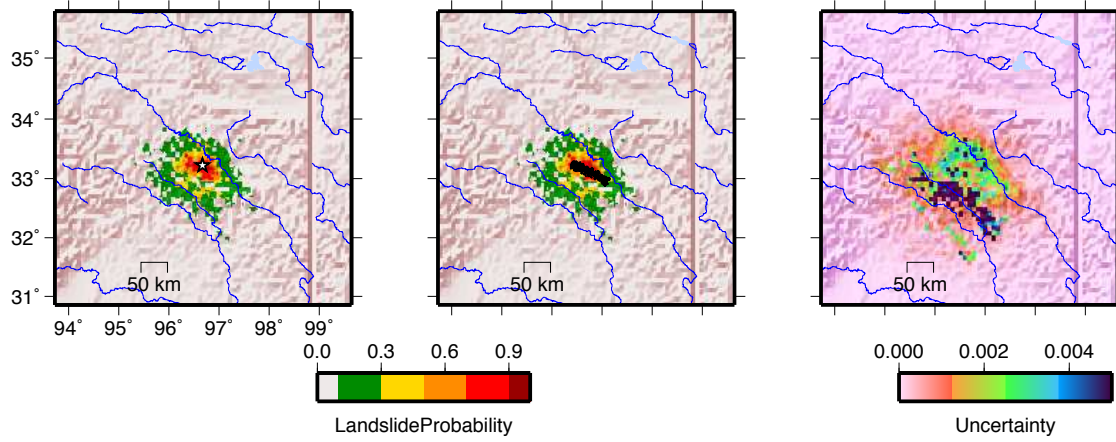


Figure S3O. Tohoku, Japan

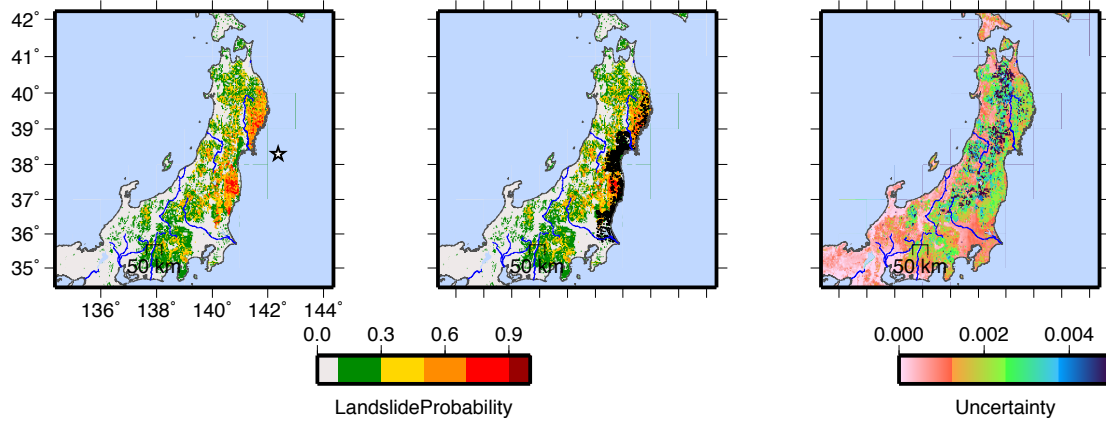


Figure S3P. Lushan, China

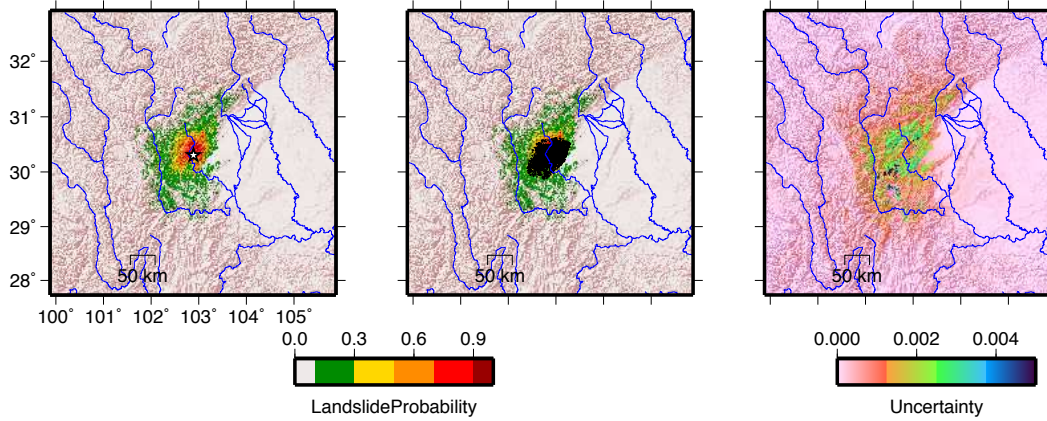


Figure S3Q. Minxian-Zhangxian, China

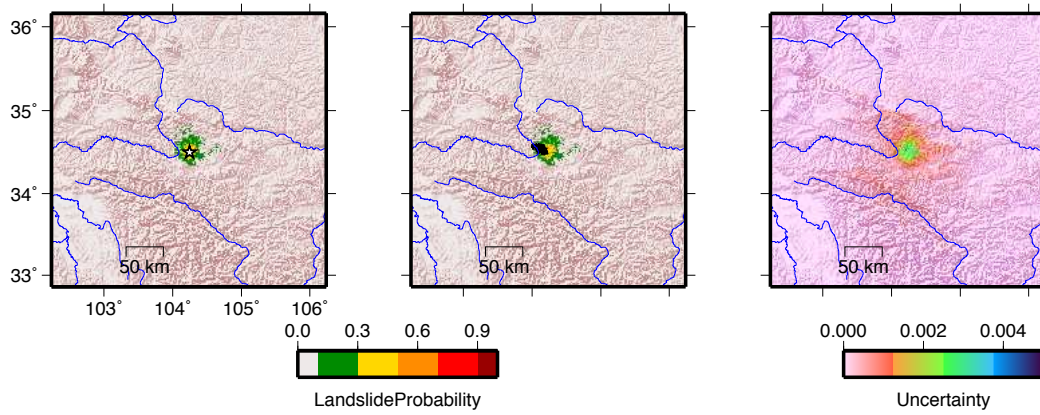


Figure S3R. Ludian, China

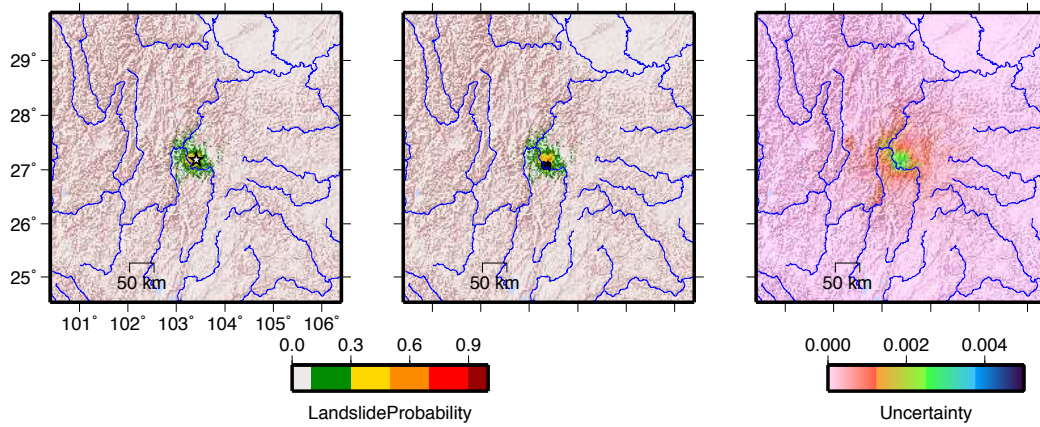


Figure S3S. Kumamoto, Japan

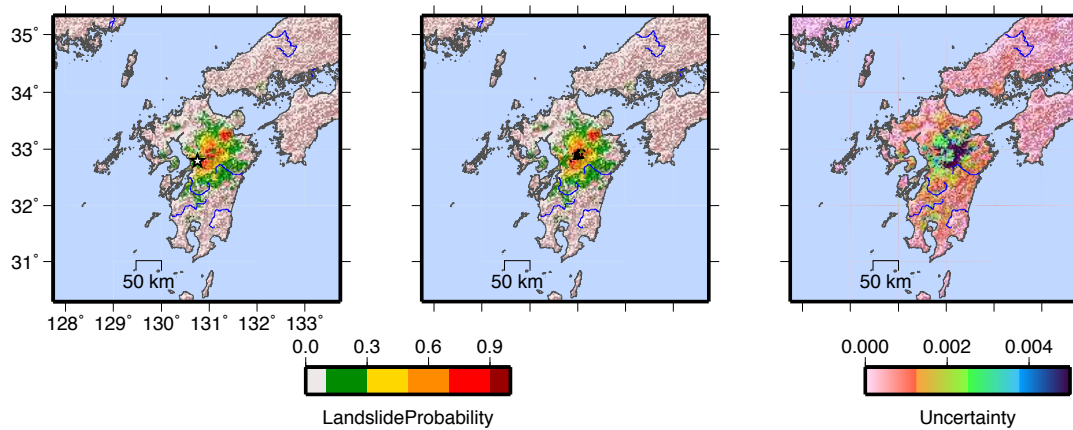


Figure S3. Model predictions in comparison to the corresponding landslide inventory for each training event.

Figure S4A. San Fernando, California

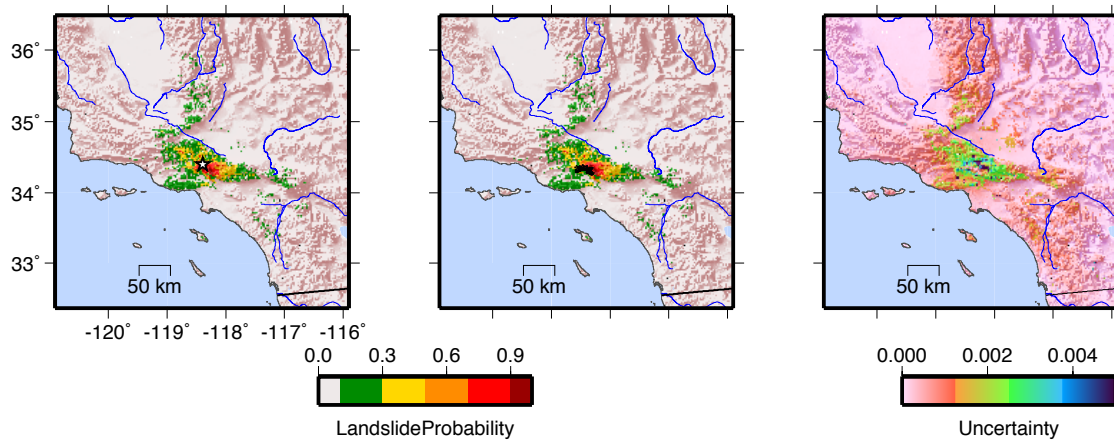


Figure S4B. San Salvador, El Salvador

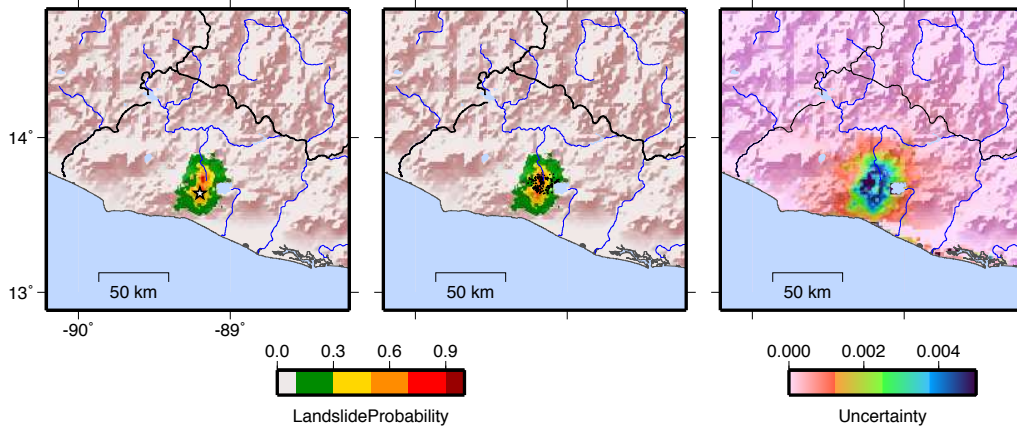


Figure S4C. El Salvador (January 2001)

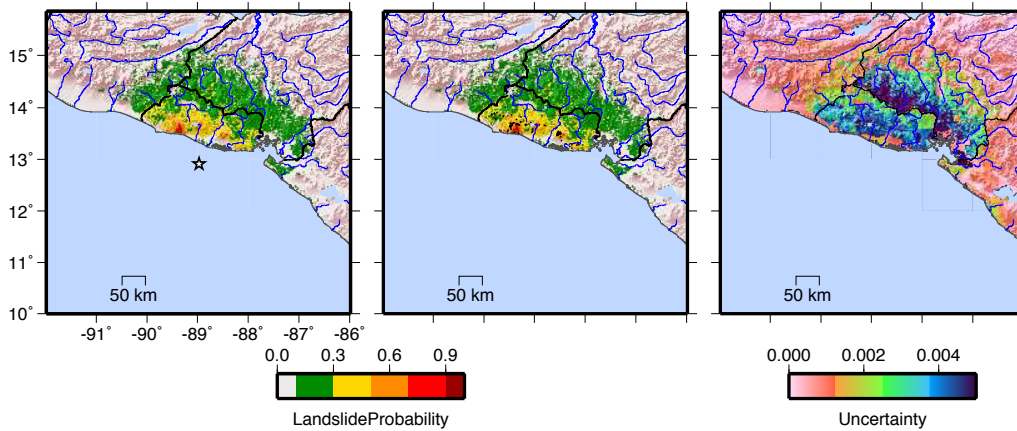


Figure S4D. El Salvador (February 2001)

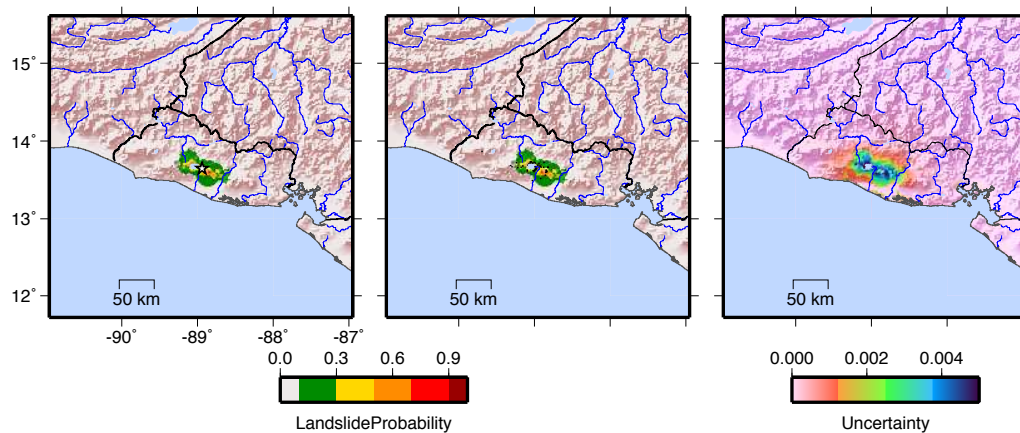


Figure S4E. Avaj, Iran

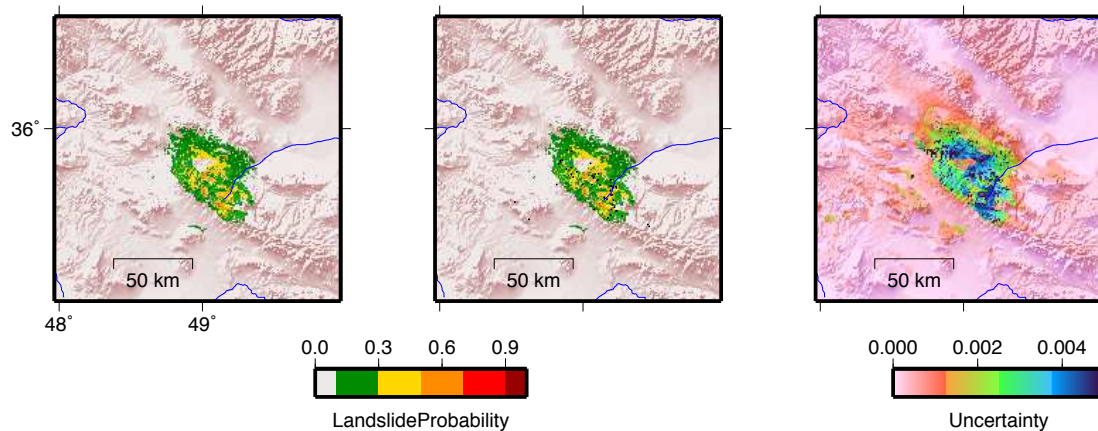


Figure S4F. Abruzzo, Italy

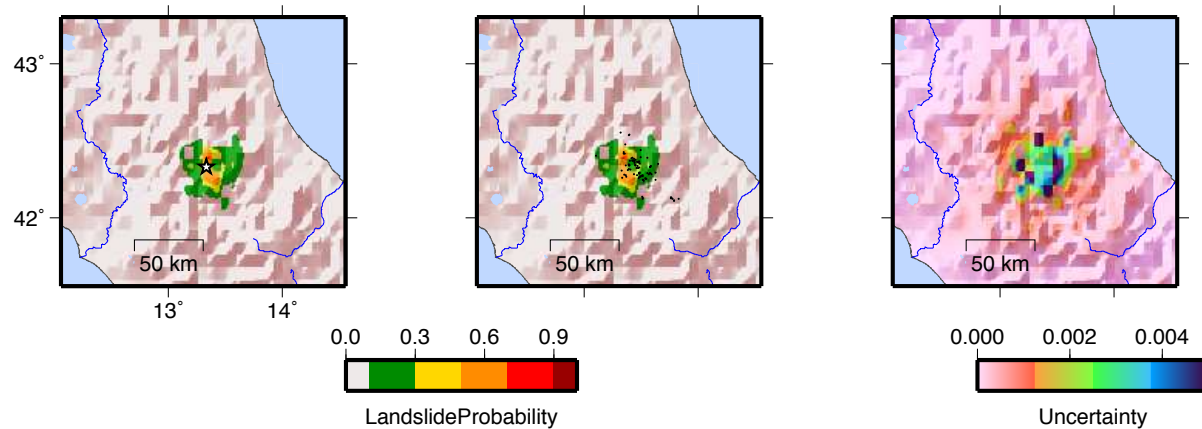


Figure S4G. Sumatra, Indonesia

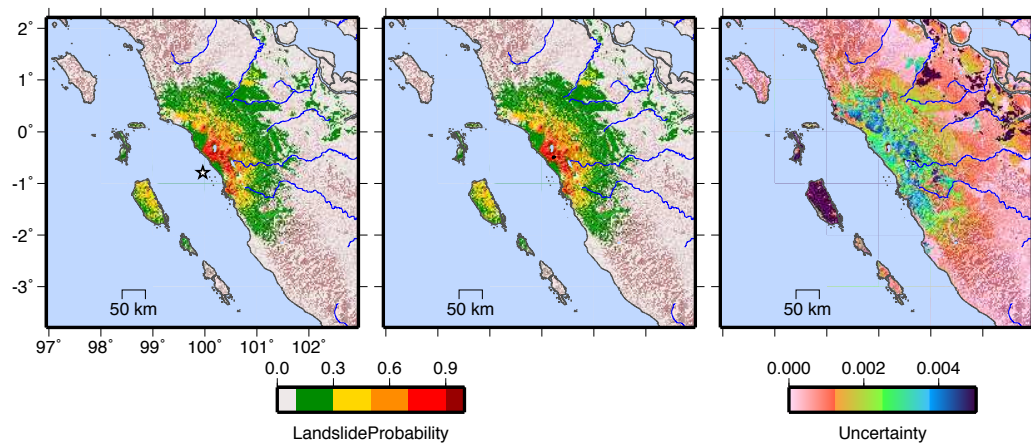


Figure S4H. Lorca, Spain

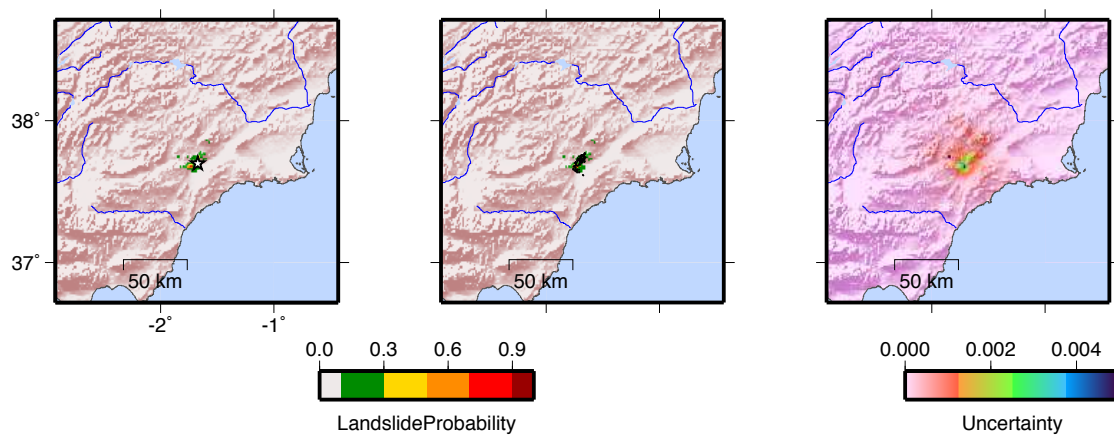


Figure S4I. Cook Strait, New Zealand

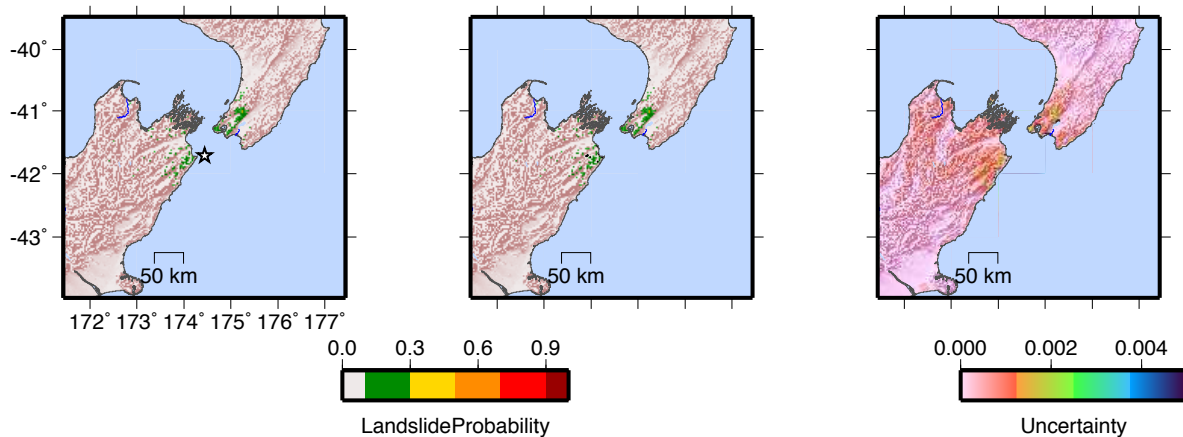


Figure S4J. Lake Grassmere, New Zealand

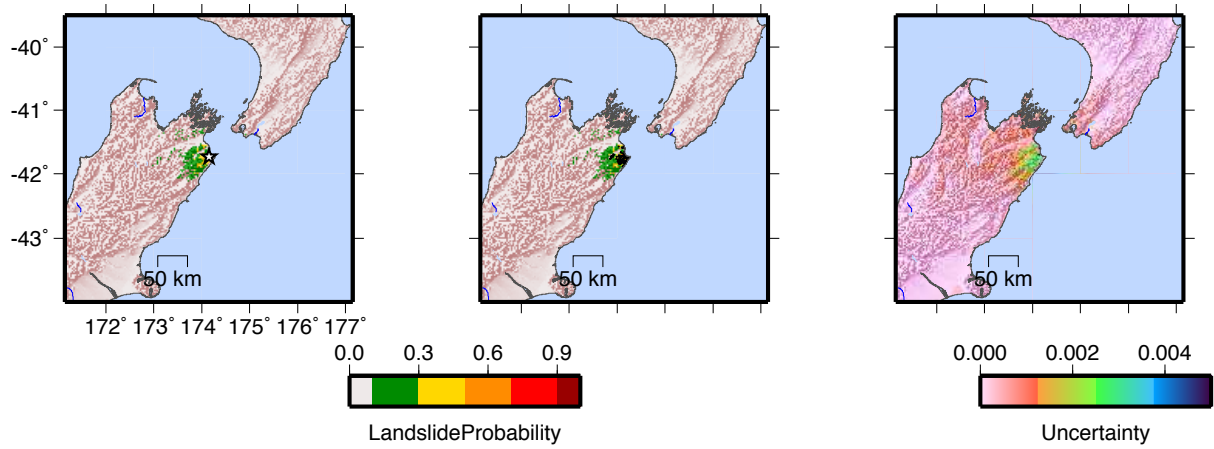


Figure S4K. Eketahuna, New Zealand

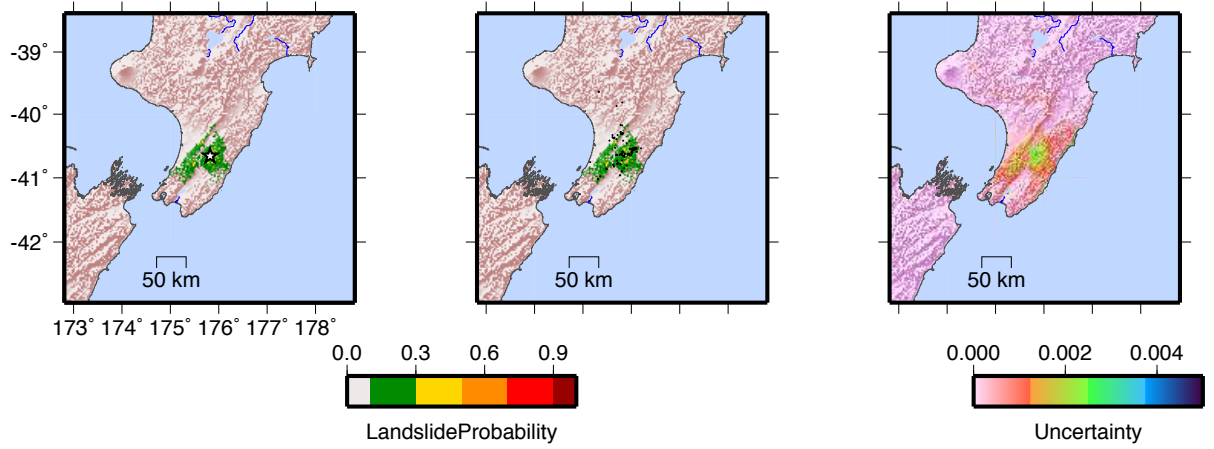


Figure S4L. Wilberforce, New Zealand

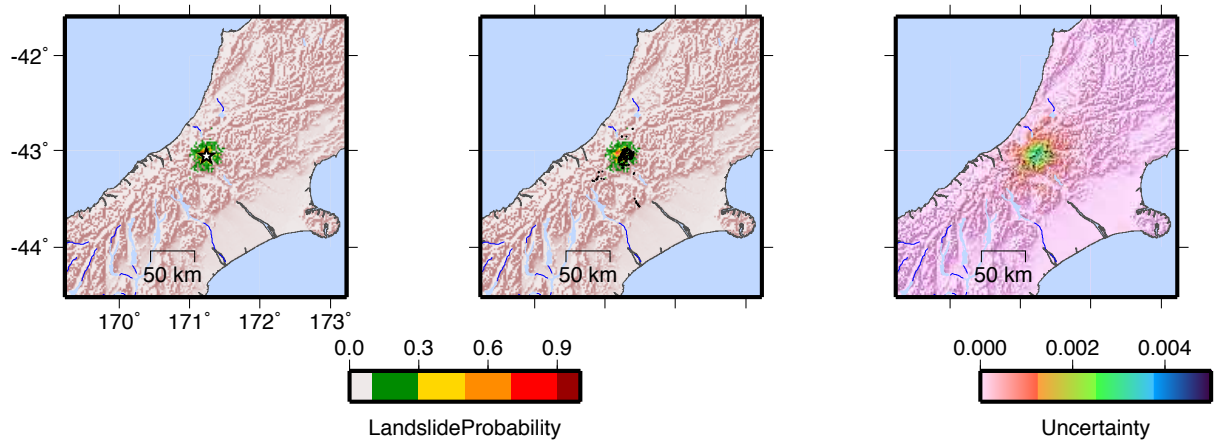


Figure S4. Model predictions in comparison to the corresponding landslide inventory for each blind validation event (note these data were not used to train the model, and are the lower scoring landslide inventories found with the data quality calculations).

	Model Intercept	Model Coefficient
PGA	-11.434	3.219
PGV	-4.860	1.756
MMI	-9.211	1.441
Slope	0.082	0.082
Max Slope	-3.914	0.103
GLIM Lithology	1.558	-0.158
Friction	-19.770	0.696
Strength	-2.619	0.742
CTI	0.512	-0.140
Globcover	-0.716	0.011
Landcover 2014	-3.694	1.090
Modis Landcover	2.340	-0.327
Modis % Green Veg	4.894	-0.052
Mean Monthly Precipitation	-1.132	0.013
Aridity	-0.818	0.000
Potential Evapotranspiration Index	0.259	-0.002
Median Elevation	0.345	0.000
Mean Elevation	0.345	0.000
Magnitude	5.249	-0.667

Table S1. Model coefficients from logistic regressions run for individual predictor variables. These correspond to the curves shown in Figure S2.

Model Formula	AIC	AUC	Accuracy	PPV	NPV	Likelihood Ratio Statistic	McFadden's R ²	Maximum Likelihood R ²	Craig & Uhler's R ²
ln(PGV)*Slope+Lithology+CTI+GlobCover	905715	0.9272	0.8508	0.8336	0.8702	957976	0.5140	0.5096	0.6795
ln(PGV)+Slope*CTI+Lithology+GlobCover	906153	0.9270	0.8508	0.8322	0.8720	957538	0.5138	0.5095	0.6793
ln(PGV)*CTI+Slope+Lithology+GlobCover	906395	0.9270	0.8509	0.8325	0.8719	957296	0.5137	0.5094	0.6792
ln(PGV)+Slope+Lithology+CTI+GlobCover	906436	0.9269	0.8509	0.8324	0.8720	957253	0.5137	0.5094	0.6792
ln(PGA)*Slope+Lithology+CTI+GlobCover	910620	0.9278	0.8545	0.8382	0.8729	953071	0.5114	0.5078	0.6771
ln(PGA)*CTI+Slope+Lithology+GlobCover	916100	0.9265	0.8543	0.8341	0.8775	947591	0.5085	0.5058	0.6744
ln(PGA)+Slope*CTI+Lithology+GlobCover	916704	0.9264	0.8541	0.8337	0.8777	946987	0.5081	0.5056	0.6741
ln(PGA)+Slope+Lithology+CTI+GlobCover	916762	0.9264	0.8542	0.8337	0.8777	946926	0.5081	0.5056	0.6741
ln(PGA)*Slope+Lithology+CTI+Percent Green Vegetation	918292	0.9266	0.8533	0.8381	0.8702	945361	0.5073	0.5050	0.6733
ln(PGA)*CTI+Slope+Lithology+Percent Green Vegetation	923667	0.9252	0.8535	0.8341	0.8758	939986	0.5044	0.5030	0.6707

Table S2. Top 10 best fitting model combinations tested, ranked by AIC value. AIC = Akaike Information Criterion; AUC = Area under the ROC curve; PPV = Positive predictive value; NPV = Negative predictive value; accuracy, PPV and NPV calculated with a 50% probability threshold for landsliding.

Table S3. Full table of model combinations tested, ranked by AIC value. AIC = Akaike Information Criterion; AUC = Area under the ROC curve; PPV = Positive predictive value; NPV = Negative predictive value; accuracy, PPV and NPV calculated with a 50% probability threshold for landsliding.

